

Granular Q-learning adaptation boosts collective welfare in multi-agent Prisoner's Dilemma

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ABSTRACT

Understanding how cooperation emerges and stabilizes in a difficult environment is a core challenge across biology, physics, and the social sciences. We present a reinforcement-learning framework for the Prisoner's Dilemma Game between the two distinct agent types: Interactive Identity (II) and Interactive Diversity (ID). While II agents compress all neighbor interactions into one strategy update, ID agents assign one strategy to each neighbor, enabling finer-grained strategic adaptation. We systematically sweep dilemma strengths and analyze both homogeneous and heterogeneous network structures to show that ID agents persistently outcompete II agents at sustaining cooperation, especially for moderate temptations to defect. Moreover, in scenarios where agents can shift from II to ID based on relative payoffs, ID learning often invades populations of II learners, though influential hub nodes can impede this transition in heterogeneous networks. Spatiotemporal analyses indicate that ID agents form a strong cluster of cooperation, which prevents defection from spreading. Finally, extrapolating these results to wider moral dimensions, such as honesty, trust, and punishment, can give a rich understanding of how this granular, neighbor-specific learning raises collective welfare within both natural ecosystems and engineered multi-agent systems.

1. Introduction

In evolutionary biology, game theory offers a unifying framework to understand how individual decision-making and social interactions can yield complex patterns of cooperation [1–3]. Within this framework, the Prisoner's Dilemma Game (PDG) stands out as a paradigmatic model capturing the conflict between individual rationality and the collective good. Although defection is often favored in pairwise interactions, many biological and social systems still exhibit robust cooperation [4–6]. This discord between theoretical expectation and empirical observation has prompted researchers to explore more nuanced mechanisms of learning, network structure, and behavioral evolution. Recent work in human–subject settings shows that the way agents gather and fuse social information can radically alter cooperative outcomes. For instance, evidence-theoretic fusion of imitation and aspiration rules [7], rational conformity learning [8], and higher-order reputation reasoning [9] all demonstrate that richer cognitive updates can lift cooperation beyond what classic imitation achieves. These findings underscore the importance of modeling learning architecture explicitly, a goal we pursue here through the II-ID contrast. In what follows we use “collective welfare” in a measurable sense: the population-average

payoff and the overall fraction of cooperative actions, both of which can be extracted directly from the simulations.

A growing body of work examines how network topology can fundamentally reshape cooperation dynamics [10–14]. Scale-free networks, for example, may facilitate cooperation through heterogeneous connectivity [15,16], whereas regular lattices or small-world networks often yield a different landscape of strategy spread [17–19]. Despite these notable developments, most existing research continues to rely on replicator-like dynamics or straightforward update rules such as imitation or proportional imitation [20,21]. Although these approaches do shed light on how network topology shapes evolutionary trajectories, they often restrict the autonomy of individual participants. As a result, agents typically end up copying the strategies of successful neighbors or adhering to fixed evolutionary prescriptions, which hinders their capacity to engage in more intricate and context-sensitive forms of adaptive learning [22,23].

Motivated by advances in machine learning, numerous studies have introduced sophisticated reinforcement learning mechanisms into evolutionary games [24–27]. In particular, Q-learning has proven effective at enabling agents to refine strategies dynamically based on accumulated rewards in a Markov decision process framework, while diverging

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from purely evolutionary models built on imitation-based updates. Yang et al. revealed that associating the state space of Q-learning agents with both neighbor strategies and payoff information can significantly enhance cooperation in spatial prisoner's dilemma settings [28], whereas Zheng et al. observed that integrating environmental information in public goods games further boosts cooperative outcomes compared to classical imitation [29]. Building on these foundations, Wang et al. showed how a particle swarm optimization-based Q-learning approach strengthens cooperation by adaptively fine-tuning exploration parameters in public goods environments [30], and Zhang et al. offered complementary evidence in the donation game, emphasizing Q-learning's ability to stabilize prosocial behavior even under strong free-riding temptations [31]. Tamura and Morita, meanwhile, corroborated these findings by replicating human-like cooperation through Q-learning agents in groups of varied sizes [32]. Despite these promising results, most Q-learning applications still treat each neighborhood's actions as a single, aggregated state, thereby overlooking the benefits of more individualized adaptation. Geng et al. partially addressed this shortcoming by illustrating how reinforcement learning can elucidate conditional cooperation patterns [33], yet an in-depth exploration of neighbor-specific decision-making remains lacking.

In a related direction, Su et al. established that interactive diversity—where each individual can behave differently with distinct neighbors—significantly enhances cooperation in both well-mixed and structured populations [34–37]. Through edge-based analyses, they demonstrated that personalized responses are instrumental in sustaining large-scale, reciprocal interactions, thereby fostering robust prosocial clusters across a broad parameter range. Gao et al. later expanded on these insights by showing that individual aspiration levels can trigger abrupt cooperative transitions in spatial prisoner's dilemma games under interactive diversity [38], reinforcing the notion that heterogeneous behaviors and adaptive thresholds can profoundly reshape evolutionary outcomes.

Building on this work, we adopt a more sophisticated learning model in which agents can operate under one of two different reinforcement learning modes. The Interactive Identity (II) agents use the aggregation of neighbor behaviors to update Q-table per timestep, whereas the Interactive Diversity (ID) agents make use of multiple iterations in order to treat each neighbor uniquely. We conjecture that ID agents, though more computationally demanding, can outperform II agents when effective reciprocity demands customized strategies for different neighbors. Our approach thus bridges ideas from Q-learning and interactive diversity, directly integrating the mathematical, physical, and life-science perspectives on how adaptive decision-making shapes biologically inspired processes of cooperation. To make the model broadly applicable, following [33], we incorporate a variation of the PDG that integrates dilemma strength theory through parameters D_r and D_g . These parameters modulate the payoff structure to interpolate between stag-hunt-like dilemmas and chicken-like dilemmas. By setting D_g equals to D_r and varying them together systematically, we map a spectrum of cooperative challenges and demonstrate how ID agents outperform II agents across diverse defection temptations. Our simulations reveal that ID agents are particularly adept at sustaining cooperation when defection is moderately attractive, thus expanding the known conditions under which prosocial outcomes can emerge.

Beyond the level of static strategy updates, we also let the possibility of agents switching type from II to ID, or vice versa, based on payoff comparisons. As an evolutionary extension, this kind of competition pushes beyond the level of strategies to include the level of learning rules. Natural systems often deal with different life organisms, groups, or AI agents applying different decision architectures, and these yielding methods might plausibly replace less adequate ones over some time. We show that this meta-level competition often favors ID agents in lattice networks but can be more nuanced in scale-free graphs, where heterogeneous connectivity may stabilize certain local interactions.

Our work synthesizes basic insights from evolutionary biology, statistical physics, and reinforcement learning in a way that is reminiscent of seminal contributions by [2,3,22]. The interplay between learning, network topology, and evolutionary dynamics is crucial for understanding cooperation both in natural systems, such as microbial communities and animal societies and in engineered systems, such as robotic swarms and distributed computational architectures. By explaining how decentralized decision-making processes can yield globally stable patterns of cooperation, we resolve a fundamental problem in many biophysical and bioengineering settings. Our in-depth investigation of interaction granularity underlines an important aspect of the adaptive capability of the agents and brings into focus the underlying mechanisms driving the emergence and persistence of complex cooperative behaviors. This unifying framework advances not only fundamental theories but also provides practical insights in designing and managing systems where cooperation is essential.

2. Methods

The evolutionary dynamics in our Prisoner's Dilemma Game (PDG) are represented as four parameters. Mainly following the setting in [33, 39], we combine the PDG with dilemma strength theory, and the modified payoff can be expressed as follows:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} = \begin{pmatrix} 1 & -D_r \\ 1 + D_g & 0 \end{pmatrix} \quad (1)$$

where $D_g = T - R$ ($0 \leq D_g \leq 1$) implies the chicken-type dilemma, and $D_r = P - S$ ($0 \leq D_r \leq 1$) represents the stag-hunt-type dilemma. Keeping $R = 1$ and $P = 0$ fixes the monetary scale; varying $(D_r, D_g) \in [0, 1]^2$ spans the full space of two-player social dilemmas. In this scaling, D_r quantifies the risk of being suckered, whereas D_g controls the extra gain available to a lone defector [39]. Extensive pilot runs showed that cooperation is far more sensitive to D_r than to D_g . A similar anisotropy is reported in [40]. We therefore follow [33] and explore the diagonal $D_g = D_r$, scanning $D_r \in [0, 0.35]$. This reduction from two dimensions to one is crucial for running 10^7 synchronous steps on lattices and scale-free graphs within a reasonable time budget. Investigating the full (D_r, D_g) plane therefore remains an interesting topic for future work.

Starting from the homogeneous networks, the agent population is initialized on an $L \times L$ grid network with periodic boundary conditions, comprising Interactive Identity (II) agents and Interactive Diversity (ID) agents in proportions governed by ρ . We first initialize the lattice by randomly assigning each node to be a cooperator or defector with equal probability. After receiving their payoffs from interactions with the four neighbors, each agent updates its Q-values. We use synchronous updating for both action selection and Q-value updates to ensure consistency across the lattice; that is, each agent's action in one round depends on the states and rewards observed in the previous round. Specifically, if an agent's action is X and its neighbor's action is Y , the payoff for that interaction is determined by the entries in Eq. (1). For instance, mutual cooperation yields $R = 1$ for each agent, whereas a cooperator facing a defector incurs $S = -D_r$ as its payoff. For II agents, they compute their payoff by aggregating the outcomes of simultaneous interactions with all neighbors. On the other hand, ID agents calculate payoffs separately for each neighbor. Lastly, agent type (II or ID) is assigned once at $t = 0$ according to the prescribed proportion ρ , whereas the initial action (C or D) is drawn independently with equal probability. From the next time step onward, actions are re-selected by the Q-learning policy, so an agent's current behavior (C/D) and its learning type (II/ID) form two independent dimensions. Consequently, we measure cooperation by the fraction of cooperative actions executed in each round, rather than by tracking static sub-populations such as 'II-C' or 'ID-D'.

We employ Q-learning as the central reinforcement learning mechanism for Interactive Identity (II) agents, allowing them to decide which action to use at each step based on past environmental observations. Q-learning is modeled as a Markov Decision Process that measures

the fitness of a particular strategy under each state by considering environmental information from the previous step. Formally, it can be defined as a tuple $\langle S, A, r, p \rangle$, where S and A represent the state and action spaces of the agent, respectively. The reward function $r : S \times A \rightarrow \mathbb{R}$ specifies the immediate reward obtained when the agent takes action $a \in A$ in state $s \in S$, and $p : S \times A \rightarrow \Delta$ is the state transition function, with Δ being the set of probability distributions over S . For an Interactive Identity (II) agent selecting action a in state s , the corresponding Q-value $Q_t(s, a)$ is updated according to

$$Q_{t+1}(s, a) = Q_t(s, a) + \alpha \left[r_t(s, a) + \gamma \max_{a' \in A} Q_t(s', a') - Q_t(s, a) \right] \quad (2)$$

where $\gamma \in [0, 1]$ is the discount factor, $\alpha \in [0, 1]$ is the learning rate, and $\max_{a' \in A} Q_t(s', a')$ denotes the maximum Q-value obtained in state s' . The symbol r_t denotes the actual reward received at time t that accumulated from all of its neighbors—i.e., the value returned by the reward function r based on the current state-action pair (s, a) .

In our model, the set of state S for Interactive Identity (II) agents is composed of the number of cooperative agents among the agent itself and his neighbors. That is, for an Interactive Identity (II) agent, the state space S_{II} is defined as the number of cooperators among the focal agent and its four neighbors. Formally, we define:

$$S_{II} = \{s \in \mathbb{Z} \mid 0 \leq s \leq 5\}, \quad (3)$$

where s represents the total count of cooperative actions observed in the local neighborhood (including the agent itself). Each state s corresponds to a column in the Q-table and is associated with two possible actions: cooperation ($a = 0$) and defection ($a = 1$). The action set A is also a strategy set, which only has two elements, namely, 0 for cooperation (C) and 1 for defection (D). We select $\alpha = 0.1$, to allow moderate updates to the Q-values each round, ensuring stability while still enabling adaptation. We set the discount factor $\gamma = 0.9$ to emphasize the importance of future rewards without completely overshadowing immediate payoffs. Finally, and $\epsilon = 0.02$ provides a low but non-negligible level of exploration, avoiding premature convergence to suboptimal policies.

At each time step, every II agent updates its 2×6 Q-table following Eq. (4), reflecting two possible actions and six local states determined by the agent's own choice plus those of up to four neighbors. Given the presence of stochasticity in action selection, each II agent employs a greedy decision-making process with exploratory deviation, this simplifies the tie handling and accelerates our large-scale GPU simulations. Specifically, the agent chooses a random strategy with probability ϵ , and selects the strategy with the highest Q-value with probability $1 - \epsilon$. Accordingly, for an agent in state s , the probability that an action a is chosen at the next time step becomes:

$$P_t(s, a) = \begin{cases} (1 - \epsilon) + \frac{\epsilon}{2} & \text{if } Q_t(s, a) \text{ is the highest,} \\ \frac{\epsilon}{2} & \text{otherwise.} \end{cases} \quad (4)$$

This scheme ensures a balance between exploitation of the current best-known action and exploration of alternative actions.

Meanwhile, for Interactive Diversity (ID) agents, they build on the Q-learning approach used by II agents but refine their actions at a more granular level. Instead of selecting a single universal action for all neighbors, an ID agent separately considers each neighbor's state and can adopt different actions for different neighbors within the same time step. For example, in a grid network where each agent has four neighbors ($n = 4$), an ID agent updates its Q-values four times per step, once for each specific neighbor. As a result, ID agents perform more frequent Q-table updates and thereby exhibit more sophisticated adaptive behavior in competitive environments. Formally, if an ID agent with n neighbors chooses action a_i in state s_i , its Q-value is updated by:

$$Q_{t+1}(s_i, a_i)$$

$$= Q_t(s_i, a_i) + \alpha \left[r_t(s_i, a_i) + \gamma \max_{a' \in A} Q_t(s'_i, a'_i) - Q_t(s_i, a_i) \right], \quad i = 1, 2, \dots, n \quad (5)$$

where α is the learning rate, γ is the discount factor, and $\max_{a' \in A} Q_t(s'_i, a'_i)$ denotes the maximal Q-value in the next state. The symbol r_t denotes the immediate reward received by the agent at time step t as a result of its interaction with a specific neighbor. It reflects the environment's feedback to the agent after taking action a_i in state s_i during its pairwise interaction at that moment. This neighbor-by-neighbor update allows ID agents to tailor their actions to each interaction, potentially conferring a significant advantage in environments where nuanced reciprocity is essential.

In our model, the state set $S_{ID}^{(i)}$ for each Interactive Diversity (ID) agent is defined for every agent-neighbor pair as:

$$S_{ID}^{(i)} = \{s_i \in \mathbb{Z} \mid 0 \leq s_i \leq 2\}, \quad (6)$$

where s_i denotes the number of cooperative agents between the focal agent and its i th neighbor. The action set A remains binary, with 0 indicating cooperation (C) and 1 indicating defection (D). We fix the parameters $\alpha = 0.1$, $\gamma = 0.9$, and $\epsilon = 0.02$, and at every time step, each ID agent autonomously updates the entries in its 2×3 Q-table according to Eq. (5), reflecting two possible actions and three possible states (determined by the agent's own choice combined with a single neighbor's action). Similar to the Interactive Identity (II) agents, a Greedy Algorithm governs the ID agent's action choices under stochastic exploration to automatically resolves ties and accelerate the GPU implementation of our million-step simulations. With probability ϵ , the agent selects a random strategy, and with probability $1 - \epsilon$, it chooses the strategy that maximizes its current Q-value. Formally, for the i th neighbor in state s_i , the selection probability is given by

$$P_t(s_i, a_i) = \begin{cases} (1 - \epsilon) + \frac{\epsilon}{2} & \text{if } Q_t(s_i, a_i) \text{ is the highest,} \\ \frac{\epsilon}{2} & \text{otherwise} \end{cases}, \quad i = 1, 2, \dots, n \quad (7)$$

This arrangement ensures that ID agents can adapt independently to each neighbor's behavior, providing them with a more nuanced decision-making capability.

We conduct simulations on two distinct types of homogeneous and heterogeneous networks: grid networks and scale-free networks. For the grid networks, each system has L^2 nodes with periodic boundary conditions, where L is set to 100, 200, 400, or 800. This configuration ensures that no finite-size effects emerge. For the scale-free networks, we generate graphs of the same size as the corresponding grid cases with an average degree of 4, consistent with the grid network. In every scenario, half of the agents initially cooperate toward their neighbors while the other half defect, regardless of whether they adopt the Interactive Identity (II) or Interactive Diversity (ID) learning scheme. Moreover, We define ρ as the initial proportion of Interactive Diversity (ID) agents in the population, with the remainder of the agent, namely, the proportion of $1 - \rho$ nodes occupied by Interactive Identity (II) agents. Unless otherwise specified, all agent nodes are randomly assigned an agent type based on this ratio at the initialization phase. Typically, each simulation runs for 10^7 time steps, and we replicate all parameter combinations at least 30 times, using different random seeds to ensure variability. To capture long-term behavior, we compute the average fraction of cooperative actions over the final 5000 time steps of each run, after transient dynamics have subsided. Specifically, since each agent simultaneously chooses an action toward each of its four neighbors at every time step, the total number of actions per time step equals four times the total number of agents in the network. We define the cooperation level as the ratio between the total number of cooperative actions taken in the final 5000 steps and the total number of actions across the network:

Cooperation Level

$$= \frac{\text{Total number of cooperative actions in the final 5000 steps}}{4 \times N \times 5000}, \quad (8)$$

where N is the number of agents. This protocol yields robust estimates of equilibrium cooperation and, if needed, allows us to compute standard errors or confidence intervals. The time-averaged value of Eq. (8), together with the contemporaneous mean payoff per agent, serves as our quantitative notion of collective welfare.

To consolidate the aforementioned components of the learning framework, Algorithm 1 provides a complete pseudocode for the simulation process executed in each round. It captures the behavioral differences between II and ID agents in terms of state construction, decision making, and Q-table updating.

Algorithm 1: Pseudocode of simulation in each round

```

1  foreach agent  $(i, j)$  in the network do
2    Determine current state based on agent type:
3    if agent is II then
4       $s_{ij} \leftarrow \#$  cooperators among self and neighbors
5    else
6       $s_{ij}^d \leftarrow \#$  cooperators with each neighbor  $d$ 
7    end
8  end
9  foreach agent  $(i, j)$  in the network do
10   Select action based on the following rule:
      • With probability  $\epsilon$ , choose action randomly;
      • Otherwise, if  $Q(C) = Q(D)$  at current state, choose randomly;
      • Otherwise, choose the action with the higher Q-value.
11   if agent is II then
12     Apply the rule once for own state  $s$ 
13   else
14     Apply the rule independently for each direction  $d$  using state  $s_d$ 
15   end
16 end
17 foreach agent  $(i, j)$  in the network do
18   Compute reward based on PD payoff matrix
19   Update Q-values based on agent type:
20   if agent is II then
21      $Q_{II}[a_{last}][s] \leftarrow$ 
22        $Q_{II}[a_{last}][s] + \alpha(r + \gamma \cdot \max Q(s') - Q_{II}[a_{last}][s])$ 
23   else
24     Accumulate updates across all directions:
25      $T[a][s] \leftarrow T[a][s] + \alpha(r + \gamma \cdot \max Q(s') - Q[a][s])$ 
26     Apply  $T$  to update  $Q_{ID}$ 
27   end
28 end

```

3. Results

In this section we explore how various Q-learning designs impact cooperative behavior and overall payoffs in networked populations. An Interactive Identity (II) agent combined the actions of all its neighbors into a single aggregate state. With this collective view, II agents were able to support cooperation in equilibrium, but could not adjust their strategies to the specific neighbor's actions. As a result, the strategy failed to capture the complex relationships that form during local interactions. To rectify this, we presented the Interactive Diversity (ID) agent, which keeps track of every neighbor's move, calculates personalized game states, and chooses different actions for each game. One of the things we want to investigate is how this more detailed learning process can enhance general cooperation and increase agent payoffs. As will be highlighted by the results, the ID agents manifest salient benefits

over II agents in maintaining robust cooperative dynamics for a much larger range of parameters setting.

Fig. 1 shows cooperation rates for three cases, exploring for the gap of Interactive Diversity versus Interactive Identity on grid networks; it is noticed from the resultant that generally speaking, ID maintains higher cooperation rates with respect to its counterpart in a wide range of most parameter sets. The level of cooperation decreases gradually and then collapses when D_r increases from 0.00 to about 0.17, while afterwards cooperation levels approach zero for all conditions, indicating that both types abandon cooperation in the face of stronger defection pressures. A direct comparison between the first only condition II agents and the third condition only ID agents shows that there is a big gap in performance: whereas ID agents at $D_r = 0.00$ get close to the perfect cooperation level of 1, II agents stabilize at around 0.8. With increasing D_r , the II agents experience a quicker decline in cooperation, while ID agents remain at a relatively high level of cooperation within the general downward trend.

In an equally divided setting with $\rho = 0.5$ —that is, half of the agents are II and the other half are ID—Fig. 2 provides a detailed breakdown of four types of pairwise interactions: II-to-II, II-to-ID, ID-to-II, and ID-to-ID, measured in stationary states. These labels are directional by design and refer to the action originating from the first agent toward the second. For example, “ID-to-II” denotes actions chosen by an ID agent directed at its II neighbor, while “II-to-ID” captures the reverse direction. Because each agent independently selects its action toward each neighbor, the bidirectional interaction between two agents yields two distinct data points: one for each direction.

Among these categories, the ID-to-ID combination consistently attains the highest cooperation levels, indicating a strong synergistic effect when both interacting neighbors employ Interactive Diversity. In contrast, the other three types yield more modest outcomes. Notably, once D_r surpasses 0.06, the ID-to-ID interaction overtakes II-to-II and II-to-ID in terms of cooperation, suggesting that even one-sided neighbor-specific adaptation from the ID agent can significantly enhance dyadic cooperation.

The observed differences between II-to-ID and ID-to-II stem from the inherent asymmetry in learning architecture. While II agents base their decision on aggregated neighbor states and apply a single action uniformly, ID agents evaluate each neighbor individually and may adjust actions more responsively. Consequently, in an II-to-ID pair, the II agent treats the ID neighbor no differently than any other neighbor, whereas in an ID-to-II pair, the ID agent can selectively respond to the II neighbor's behavior. This asymmetry leads to measurable differences in cooperation outcomes, validating the need to analyze these directions separately.

We present in Fig. 3 the dynamics of cooperation for D_r between 0.00 and 0.3 over 10^7 steps of simulation. Every panel reflects a different composition of agents: $\rho = 0$ (Fig. 3(a)) is composed only of II agents, while $\rho = 1$ (Fig. 3(b)) is composed only of ID agents. We plot 30 individual curves per scenario, highlighting three key D_r values (0.03, 0.15, and 0.3) to facilitate direct comparisons of cooperation trajectories. In Fig. 3(a), for example, all of the cooperation levels start at 0.5, but different D_r values result in divergent dynamics: $D_r = 0.3$ rapidly collapses to near-zero cooperation by the 10^4 th round, whereas $D_r = 0.03$ eventually rises to a stable 0.7. Moreover, the curve of the $D_r = 0.15$ case decays slower and saturates near 0.3, indicating a state of partial cooperation rather than complete collapse. On the other hand, for the ID agents, Fig. 3(b) plots larger fluctuations against very similar critical threshold values of D_r at the high end. Especially, the curve for $D_r = 0.3$ always declines to zero, whereas that for $D_r = 0.15$ sinks to around 0.3 at around the 10^4 th round and finally stabilizes and goes back to approximately 0.7 late in the simulation. Taking these together, the following results show how temptation strength dictates the long-term cooperation patterns of homogeneous populations of either II or ID agents, and that even small differences in D_r can make large changes in the stability and eventual level of cooperation within each agent type.

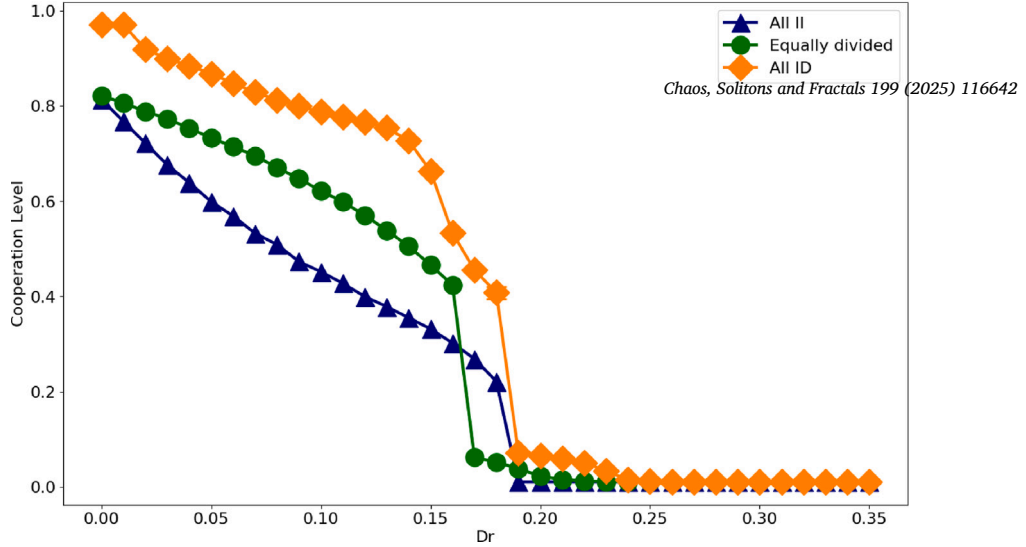


Fig. 1. Final cooperation level by D_r with three agent proportions (ρ) on a grid network. This figure displays the final cooperation fraction for $\rho = 0$, $\rho = 0.5$, and $\rho = 1$, where ρ denotes the proportion of Interactive Diversity (ID) agents in a population otherwise composed of Interactive Identity (II) agents. The agents occupy a 400×400 grid with periodic boundaries and interact with their four nearest neighbors. We vary the defection temptation parameter D_r from 0.00 to 0.35 in increments of 0.01, while other parameters $\alpha = 0.1$, $\gamma = 0.9$, and $\varepsilon = 0.02$ remain fixed. After 10^7 simulation steps, the cooperation level is calculated by averaging over the final 5000 steps. We perform 30 independent trials for each scenario, and the standard errors are proportional to the markers. The figure highlights how increasing D_r eventually drives cooperation toward zero, but the rate and threshold of decline depend strongly on the proportion ρ of ID agents.

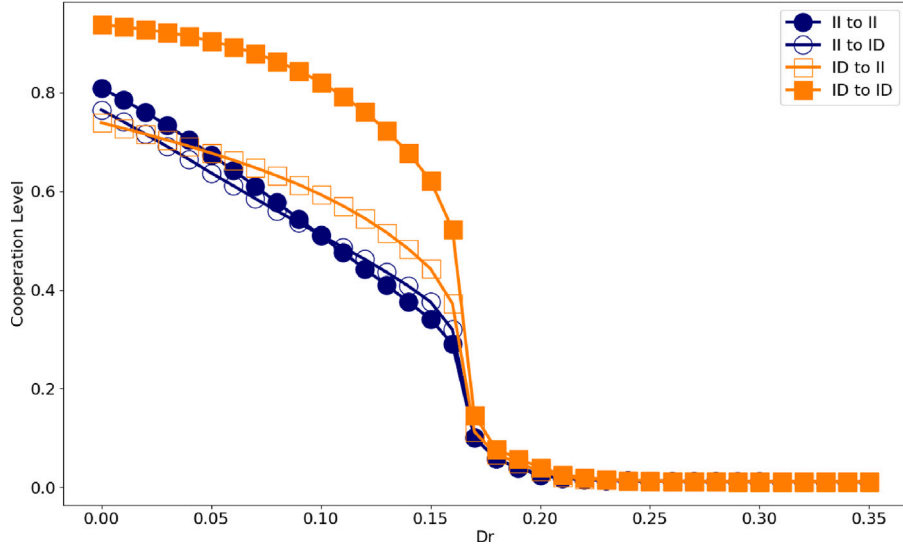


Fig. 2. Final cooperation level by different types of agents in a grid network with $\rho = 0.5$. Here, $\rho = 0.5$ implies an equal mix of Interactive Identity (II) agents and Interactive Diversity (ID) agents initialized on a 400×400 grid with periodic boundaries. Four pairwise interaction types are examined: II-to-II, II-to-ID, ID-to-II, and ID-to-ID. The cooperation level is plotted against D_r , ranging from 0.00 to 0.35. After 10^6 simulation steps, the cooperation level is calculated by averaging over the final 5000 steps. We perform 30 independent trials for each scenario, and the standard errors are proportional to the markers. Results show that ID-to-ID interactions sustain the highest cooperation, while other pairwise types yield more modest outcomes. These trends underscore how neighbor-specific learning (ID) can promote robust cooperative clusters relative to aggregated strategies (II).

Next, turning to Fig. 4, we extend the analysis of Fig. 3 by showing the spatio-temporal distribution of cooperative and defective actions for three representative values of D_r , namely 0.03, 0.15 and 0.3. Panel (a) reflects $\rho = 0$ where II agents are either “cooperative” or “defective” according to their chosen actions. Early in the simulation, the cooperative and defective strategies are roughly balanced, with equilibrium maintained to the 10^3 rd round. Between the 10^3 rd and 10^4 th rounds, a dramatic transition occurs. For $D_r = 0.03$, clusters of cooperators develop steadily and eventually win at the end of the run. In contrast, $D_r = 0.15$ and $D_r = 0.3$ both result in the dominance of defectors over time, with cooperation reduced to scattered patches at $D_r = 0.3$. While $D_r = 0.15$ sustains a fair number of cooperative clusters, these are smaller and more fragmented, yielding a stable

but lower level of overall cooperation. For panel (b), which presents the case of $\rho = 1$, we further classify each ID agent’s behavior into five categories—most cooperative, cooperative, neutral, defective, and most defective—based on the number of cooperative actions directed toward its four neighbors. Specifically, agents that cooperate in all four directions are labeled as most cooperative, while those cooperating with three, two, one, or zero neighbors are categorized as cooperative, neutral, defective, and most defective, respectively. This rule-based classification allows us to visualize the spatial distribution of fine-grained behavioral tendencies across the network. Here, agents labeled as “neutral” dominate under all values of D_r during the early stage, though a substantial pattern shift occurs after approximately 10^2 rounds. For $D_r = 0.3$, “most defective” agents rapidly spread, eventually

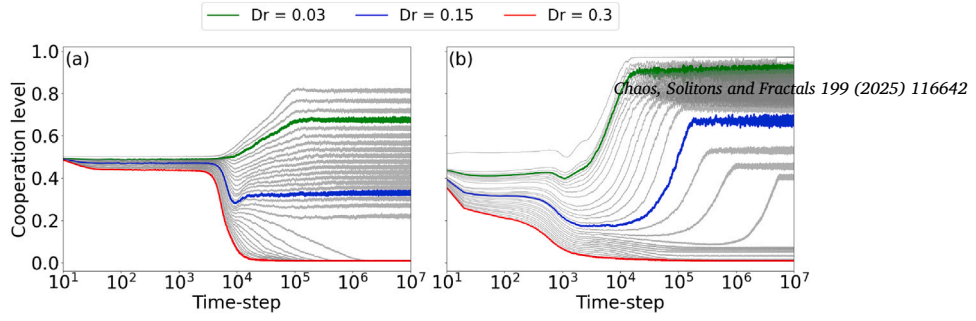


Fig. 3. Temporal evolution of cooperation levels in a grid network, highlighting the impact of D_r from 0.00 to 0.3 over 10^7 steps. Panels (a) and (b) show the cooperation trajectories under two extreme compositions: $\rho = 0$ (all II agents) in panel (a), and $\rho = 1$ (all ID agents) in panel (b). We sample 30 individual runs for each scenario, emphasizing three representative values of D_r : 0.03, 0.15, and 0.3. Additionally, both panels display 31 cooperation-level curves, corresponding to D_r values increasing from 0.00 (top) to 0.30 (bottom) in increments of 0.01. The figure reveals how moderate D_r levels allow some agents to maintain cooperation, whereas higher temptations (e.g., $D_r = 0.3$) drive widespread defection. Notably, ID agents exhibit more complex and variable trajectories, suggesting enhanced adaptability compared to II agents.

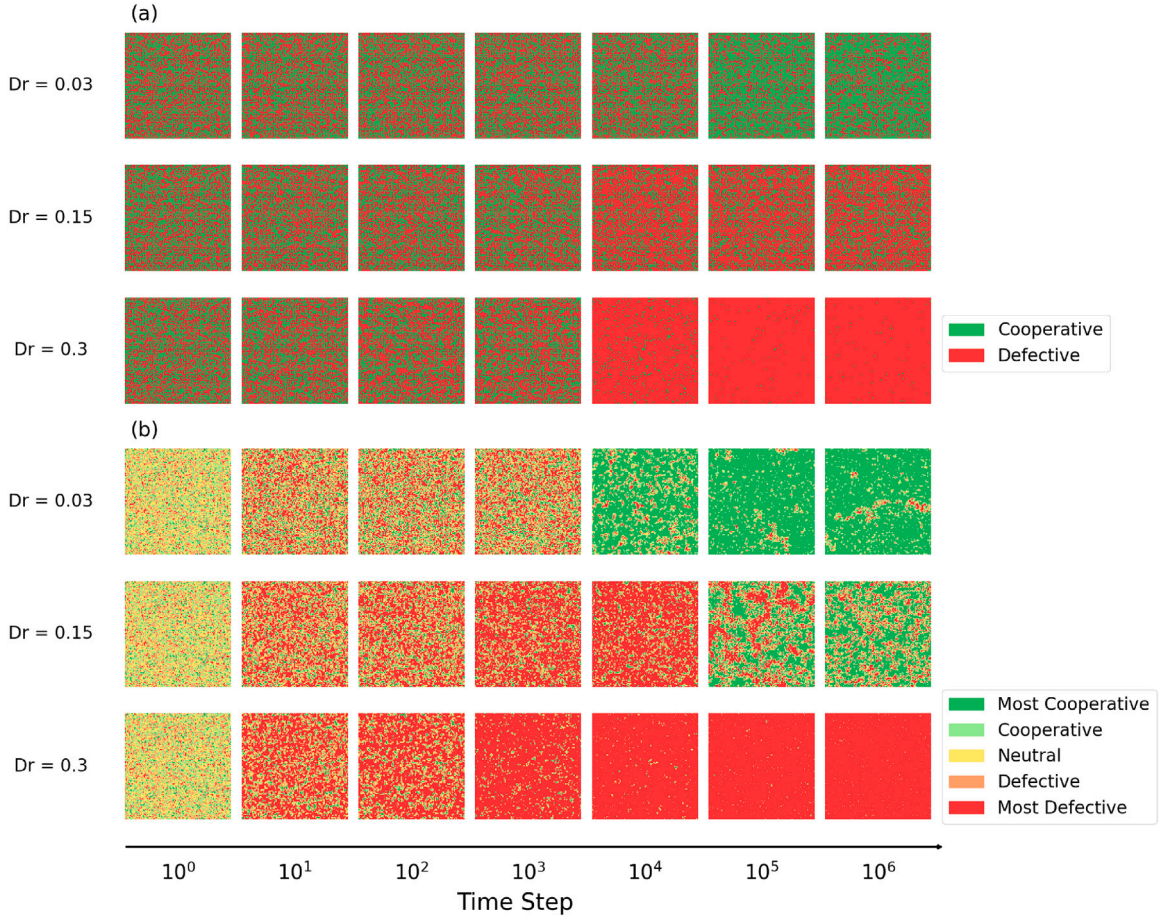


Fig. 4. Spatiotemporal action distributions on a grid network for three representative D_r values (0.03, 0.15, and 0.3). Panels (a) and (b) respectively correspond to $\rho = 0$ (all II agents, adopting a binary strategy of “cooperative” or “defective”) and $\rho = 1$ (all ID agents, capable of selecting four neighbor-specific actions and categorized into five behavioral types—most cooperative, cooperative, neutral, defective, and most defective—based on the number of cooperative actions directed toward their four neighbors). These snapshots, taken after repeated interactions and Q-learning updates, highlight a 100×100 region of the full lattice, visualizing how local decision-making gives rise to emergent cooperation or defection clusters. For $D_r = 0.03$, cooperative clusters grow and often dominate by the end of the simulation. In contrast, $D_r = 0.15$ and $D_r = 0.3$ induce higher levels of defection, resulting in fragmented or vanishing cooperative regions. These results illustrate how varying defection temptations influence spatial dynamics under homogeneous II- and ID-agent populations.

leaving only a few neutral agents by the end of the simulation. For $D_r = 0.15$, the initially expanding population of “most defective” agents is overtaken by a reversal trend, leading to a network increasingly governed by “most cooperative”, “cooperative”, and “neutral” agents after the 10^4 th round. Lastly, for $D_r = 0.03$, cooperative clusters continue to expand throughout the simulation, eventually occupying nearly the entire lattice. These findings highlight the critical role of

defection pressure in shaping the overall spatiotemporal patterns of cooperation.

We explore how three interaction categories—cooperative to cooperative (CC), cooperative to defective & defective to cooperative (CD), and defective to defective (DD)—evolve over time in Fig. 5. Panels (a) and (b) correspond to $D_r = 0.03$, where CD interactions dominate for the first 10^3 to 10^4 rounds. After this period, CC interactions surge

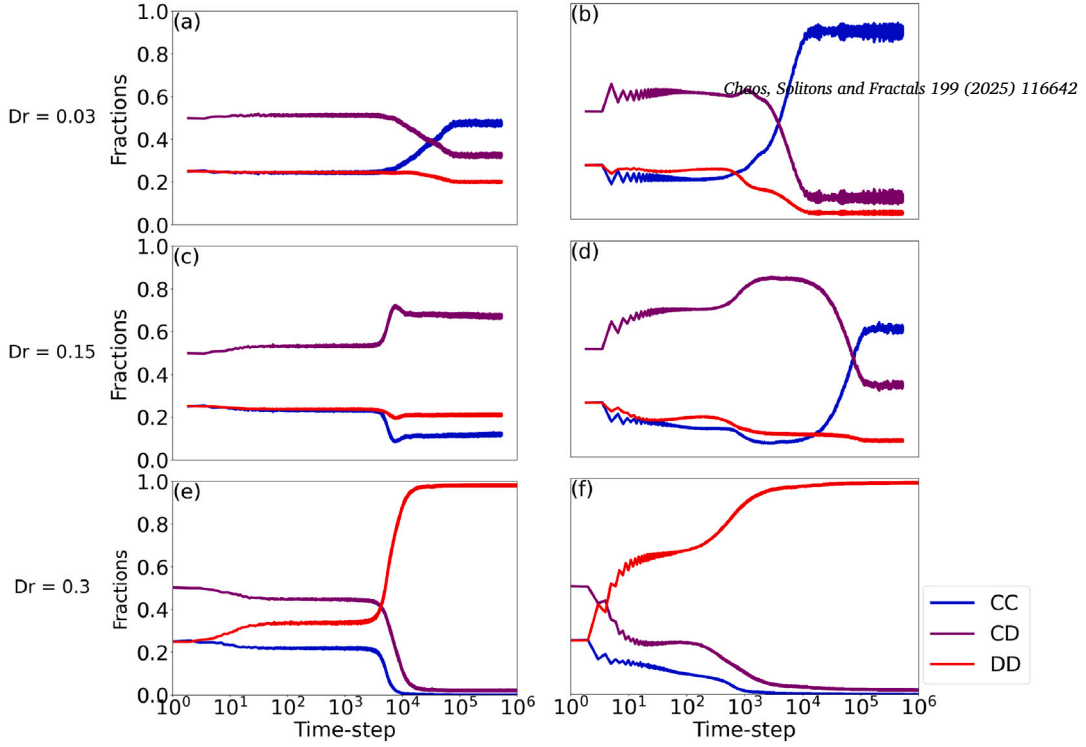


Fig. 5. Agent interaction dynamics over time in a grid network. Building on the temporal patterns shown in Fig. 3, this figure tracks how pairwise interaction types evolve for three representative D_r values: 0.03, 0.15, and 0.3. Panels (a), (c), and (e) depict the all-II-agent condition ($\rho = 0$), whereas panels (b), (d), and (f) reflect the all-ID-agent condition ($\rho = 1$). Each edge in the grid is classified as one of three categories based on the actions of its two connected agents: cooperative-to-cooperative (CC), cooperative-to-defective & defective-to-cooperative (CD), or defective-to-defective (DD). By recording these proportions over 10^6 time steps, we highlight how defection temptation (D_r) and agent type (II vs. ID) shape collective interaction patterns. All agents use Q-learning updates ($\alpha = 0.1$, $\gamma = 0.9$, and $\epsilon = 0.02$) and synchronize their decisions with four grid neighbors each round.

and eventually become the majority type. For $D_r = 0.15$, panels (c) and (d) reveal a more intricate pattern. While CD interactions remain most frequent in panel (c), they are overtaken by CC in panel (d), implying stronger cooperation among Interactive Diversity (ID) agents in that configuration. At $D_r = 0.3$, panels (e) and (f) illustrate how DD interactions overwhelm both CC and CD, aligning with the high-defection scenarios observed in Fig. 4. These findings underscore that higher defection pressure leads to a decisive tilt toward DD interactions, especially among ID agents who can amplify defectors' influence when collaboration becomes less rewarding.

Next we visualize how agent proportions evolve on a grid network with $\rho = 0.5$ and $D_r = 0.15$ in Fig. 6. Here, a type-learning mechanism allows each agent to adopt or abandon Interactive Identity (II) or Interactive Diversity (ID) strategies based on neighbor payoffs. The decision process follows the Fermi function [41],

$$W_{S_x \leftarrow S_y} = \frac{1}{1 + \exp\left(\frac{P_x - P_y}{K}\right)}. \quad (9)$$

where P_x and P_y are the payoffs of agents x and y , respectively, and $K = 0.1$ captures uncertainty in strategy uptake [42]. During the initial 10^2 rounds, the proportion of II agents experiences a mild increase, indicating early local stability. After approximately the 10^3 rd round, however, ID agents begin to outpace II agents and ultimately dominate the network. Once established, this dominance persists into the steady state, signifying the robust propagation of more adaptive learning under ID strategies. These results emphasize how an agent's capacity to switch learning modes can reshape the population composition and further cement the advantages of ID approaches when $D_r = 0.15$.

It is crucial to understand how agent-level rewards and local segregation patterns influence cooperation, therefore, we examine Figs. 7 and 8. Fig. 7 demonstrates the average reward per round across 10^6 simulation steps under $\rho = 0.5$. From $D_r = 0.00$ to 0.16, Interactive Diversity (ID) agents consistently secure higher rewards than Interactive

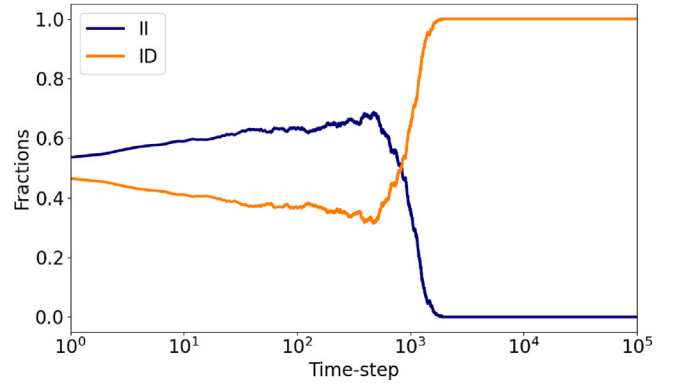


Fig. 6. Temporal evolution of agent proportions in a grid network with type-learning. This figure illustrates how agents switch between Interactive Identity (II) and Interactive Diversity (ID) strategies over time, using a Fermi-type update rule to adopt the more successful learning mode. The simulation is run on a grid of 200×200 nodes with periodic boundaries, under parameters $\rho = 0.5$ (initially half II, half ID) and $D_r = 0.15$. Those with higher payoffs are more likely to attract neighboring agents to their learning type. The plot shows how the system transitions from a mixed state to eventual dominance by the ID strategy, illustrating the competitive dynamics of Q-learning protocols within spatial populations.

Identity (II) agents, reflecting the benefit of a more adaptive learning process. However, when D_r surpasses 0.16, the growing temptation to defect diminishes this advantage, eventually driving both agent types toward similar payoff levels. Turning to Fig. 8, we investigate how segregation level η evolves at two different distances in the network—1 and 2—under the extreme conditions $\rho = 0$ (all II agents) and $\rho = 1$ (all ID agents), with D_r set to 0.15 in both cases. For II agents, cooperators are assigned a value of 1 and defectors a value of 0, so the segregation

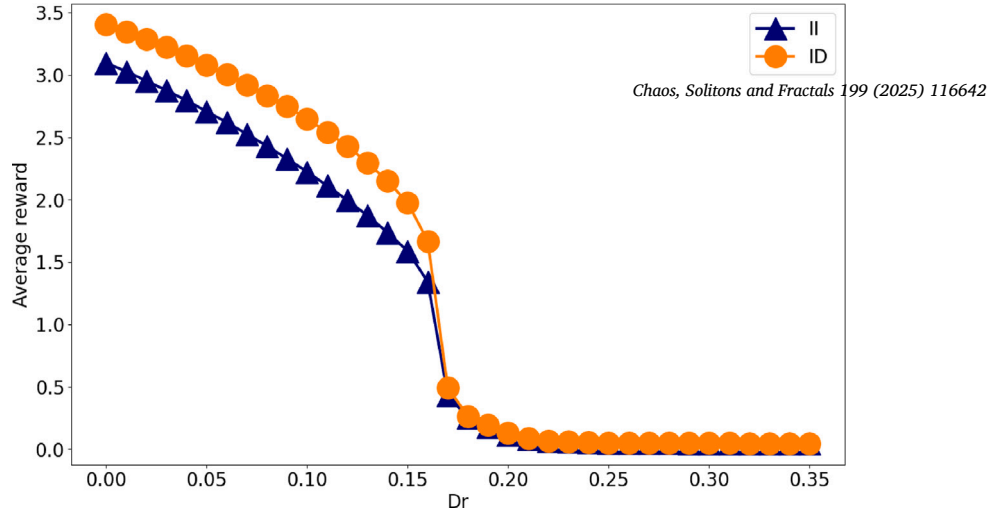


Fig. 7. Average reward analysis for II and ID agents coexisting on a grid network. This figure compares the mean payoff per agent per round for Interactive Identity (II) agents and Interactive Diversity (ID) agents coexisting at $\rho = 0.5$ in a grid environment. Each data point represents the average reward over 10^6 time steps, with the last 5000 steps used to compute a steady-state measure. We vary D_r from 0.00 to 0.35 (in increments of 0.01), while holding other Q-learning parameters constant ($\alpha = 0.1$, $\gamma = 0.9$, and $\epsilon = 0.02$). The results underscore that ID agents typically enjoy higher payoffs than II agents, thereby illustrating how granular adaptation improves the collective welfare metric based on mean payoff.

score is the absolute difference between each agent's action and the mean action of its distance 1 or 2 neighborhoods. By contrast, ID agents calculate segregation level *eta* by comparing their own number of cooperative edges towards that of their neighbors, capturing a more granular perspective. Panel (a) shows that II agents sustain a higher segregation level at distance 1 throughout the run. In panel (b), II agents also start higher at distance 2 but are quickly overtaken by ID agents, who experience a sharp rise between the 10^2 nd and 10^3 rd rounds. After a mild dip from this peak, ID agents stabilize at a higher level of segregation, whereas II agents exhibit only a modest fluctuation before settling near 0.43. These observations underscore the dynamic influence of agent type on local interaction patterns over varying spatial scales.

To place the forthcoming scale-free results in context, we first recall the grid-network curves in Fig. 1. Those curves show a sharp collapse of cooperation once D_r exceeds roughly 0.17. Fig. 9 repeats the same experiment on a scale-free network with the same average degree of 4, thereby revealing how network heterogeneity changes the landscape: the decline in cooperation is more gradual, and for intermediate temptations ($D_r \approx 0.10$ – 0.25) the scale-free population retains noticeably higher cooperation than its grid counterpart. Specifically, in each round of interaction, an Interactive Identity (II) agent computes its payoff by summing the outcomes of simultaneous interactions with all of its neighbors, while an Interactive Diversity (ID) agent evaluates payoffs separately for each neighbor based on individual interactions. Importantly, all payoffs are computed and applied within the same time step, without accumulation across rounds. This harmonized protocol ensures that any difference between Figs. 1 and 9 is attributable to topology rather than implementation. Three conditions are considered: $\rho = 0$ (only II agents), $\rho = 0.5$ (a mix of both types), and $\rho = 1$ (only ID agents). Across all tested D_r values, ID agents display higher cooperation than II agents, underlining the power of neighbor-specific learning in complex, hub-dominated networks. In such environments, node connectivity and degree distributions amplify the capacity of ID agents to sustain cooperation. Continuing this thread, Fig. 10 visualizes the equilibrium distribution of strategies in the same scale-free network, following 10^6 simulation rounds. The network maintains an average degree of 4, consistent with the grid network. As before, the payoff calculation follows the same rules: in each round, II agents compute their payoff by aggregating the outcomes of simultaneous interactions with all neighbors, whereas ID agents calculate payoffs

separately for each neighbor. All payoffs are computed within the same time step, and no payoff information is carried over to subsequent rounds. This setup ensures a consistent basis for comparing strategic dynamics between the two agent types. Panel (a) shows the $\rho = 0$ scenario with II agents, where strategies are simply “cooperative” or “defective”. After 10^6 synchronous steps at $D_r = 0.15$ the $\rho = 0$ scale-free population stabilizes at approximately 18% cooperators. These surviving *C*-nodes appear in two geometric niches: low-degree vertices, where the payoff difference between cooperating and defecting is too small to create a decisive Q-value gradient, and very high-degree hubs with $k \gtrsim 30$, where each unilateral defection alters only a negligible fraction of the many dyadic payoffs so the average advantage of *D* is heavily diluted, allowing the hub's Q-table to lock into cooperation. The co-existence of these peripheral and hub-level refuges accounts for the observed low, yet persistent, plateau of cooperation. In contrast, panel (b) illustrates the $\rho = 1$ scenario, in which ID agents can be “most cooperative” (over 80% of actions are cooperative), “cooperative” (60% to 80% of actions are cooperative), “neutral” (40% to 60% of actions are cooperative), “defective” (20% to 40% of actions are cooperative), or “most defective” (under 20% of actions are cooperative). Notably, “most cooperative” agents dominate, while strongly defecting categories remain sparse. One can see that “most cooperative” agents are dominant while strongly defecting categories are sparse. This result underlines the adaptiveness of ID agents, since their flexible, neighbor-focused decision-making manner favors cooperation even in highly heterogeneous networks, where influential hubs might otherwise enforce defection.

Focusing next on Fig. 11, we explore how cooperation correlates with node degree in a scale-free network. Panel (a) presents results for Interactive Identity (II) agents, revealing a lower baseline cooperation level compared to Interactive Diversity (ID) agents. Nevertheless, the cooperation rate for II agents steadily rises with the number of neighbors, suggesting that higher connectivity partially compensates for their simpler learning approach. In Panel (b), the ID-agent population consistently achieves higher cooperation levels relative to II agents, and the upward trend with degree is more pronounced. Consequently, these findings underscore that an agent's connectivity strongly shapes its propensity to cooperate: more highly connected nodes are more likely to sustain cooperative strategies, regardless of whether they employ II or ID mechanisms.

To contrast the network-dependent outcome of the same Fermi-rule adoption process, we track in Fig. 12 how the proportions of II

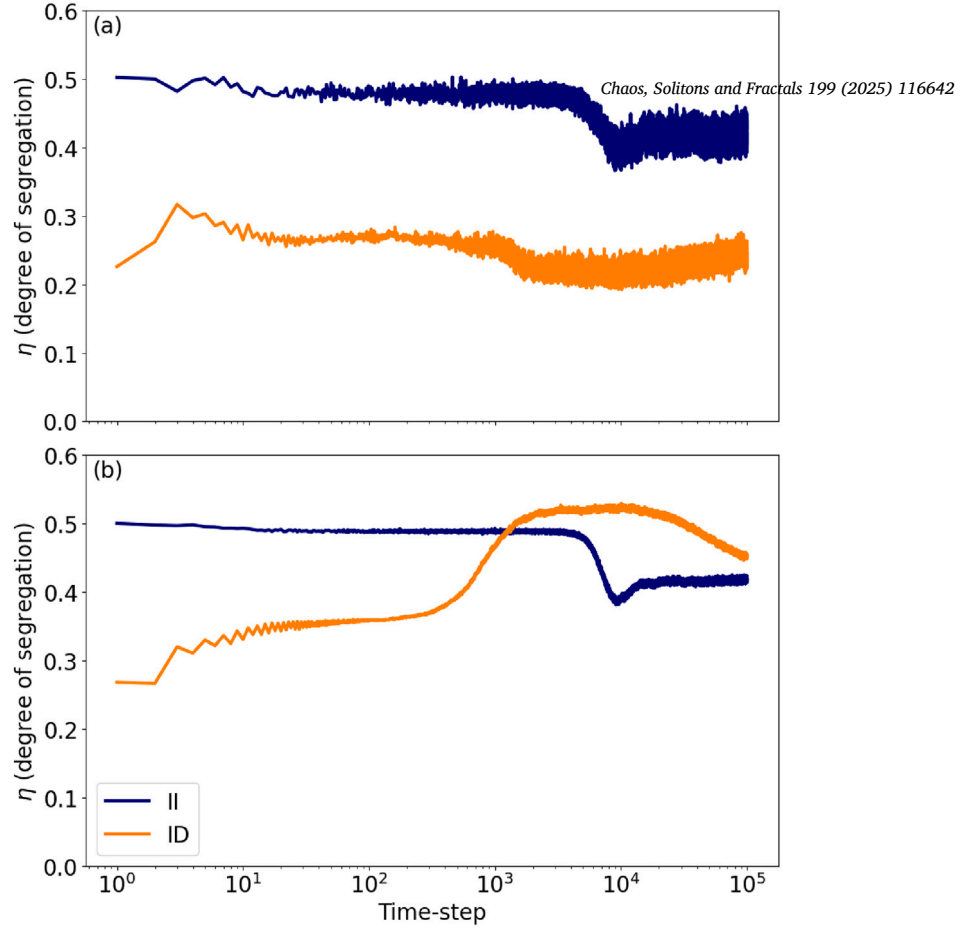


Fig. 8. Degree of segregation in a grid network at distances 1 and 2. Depicting agent-level differentiation, this figure measures how distinct each agent's action is from its neighbors' actions under two extreme scenarios: $\rho = 0$ (all II agents) and $\rho = 1$ (all ID agents). Panels (a) and (b) track segregation degrees at distances 1 and 2, respectively, over 10^6 time steps on a 200×200 grid under $D_r = 0.15$. The figure highlights how early fluctuations, mid-run spikes, and eventual equilibria emerge as agents repeatedly learn and update strategies via Q-learning.

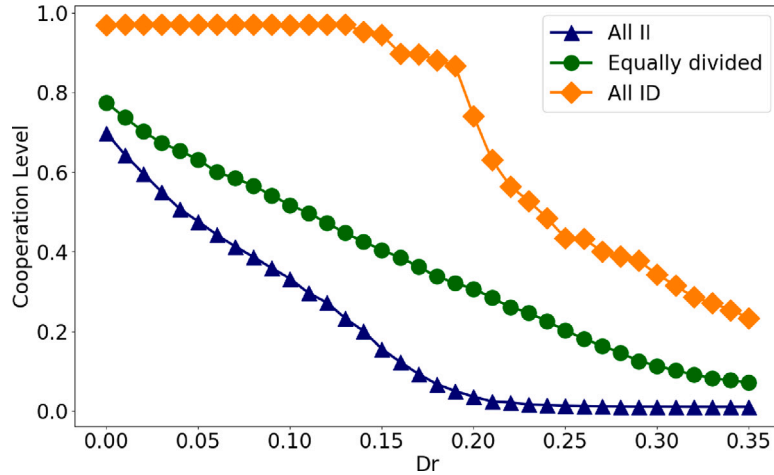


Fig. 9. Final cooperation level by D_r with three agent proportions (ρ) on a scale-free network. This figure displays the final cooperation fraction for $\rho = 0$, $\rho = 0.5$, and $\rho = 1$, where ρ denotes the proportion of Interactive Diversity (ID) agents in a population otherwise composed of Interactive Identity (II) agents. The agents occupy a scale-free networks with 40,000 nodes. Other setting is the same as Fig. 1. The figure highlights how increasing D_r eventually drives cooperation toward zero, but the rate and threshold of decline depend strongly on the proportion ρ of ID agents.

and ID agents evolve on a scale-free network ($\rho = 0.5$, $D_r = 0.15$). Although the update rule is identical to that used on the square lattice, the heterogeneous degree distribution now reverses the competitive balance: high-degree hubs accumulate payoff more rapidly under the

simpler II scheme, thereby attracting adjacent ID agents to switch. Consequently, II agents secure an early and persistent dominance in heterogeneous networks, whereas ID agents had invaded and prevailed on the homogeneous grid. This comparison underlines that it is the

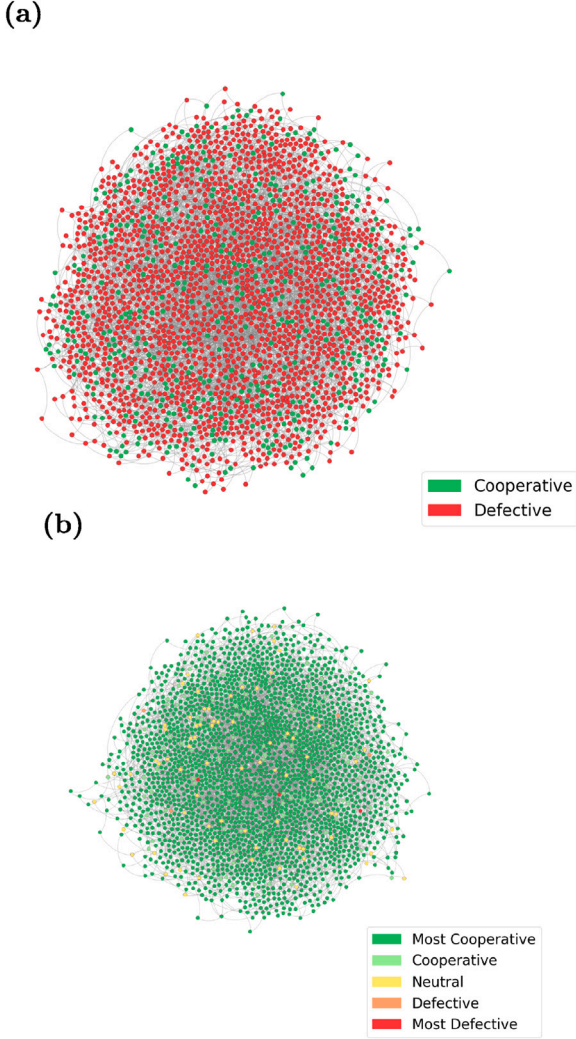


Fig. 10. Action distributions in a scale-free network of 2500 nodes at $D_r = 0.15$. We visualize the distribution of actions among II or ID agents on a smaller scale-free network (2500 nodes) for better interpretability. Panel (a) corresponds to $\rho = 0$ (all II), where cooperation stabilizes at approximately 18%—mainly among very low-degree nodes and a few high-degree hubs where the payoff gradient toward defection is too weak to overturn their initial cooperative preference. Panel (b) shows the scenario $\rho = 1$ (all ID), where an agent’s level of cooperativeness can range from “most cooperative” (over 80% of actions are cooperative), “cooperative” (60% to 80% of actions are cooperative), “neutral” (40% to 60% of actions are cooperative), “defective” (20% to 40% of actions are cooperative), or “most defective” (under 20% of actions are cooperative), as determined by neighbor-specific strategies. For visual detail of the five-color coding, see the enlarged sub-regions in Appendix Figure A.3., where every behavioral class is visible. Both panels capture the network state after 10^6 steps, highlighting how a modest defection temptation ($D_r = 0.15$) yields near-complete dominance of cooperators in the ID condition but pervasive defection in the II condition. This contrast underscores the adaptive advantage of ID agents, who can customize actions for each neighbor in a highly connected environment, also leads to a better level of cooperation.

topology—not the adoption rule itself—that determines which learning mode ultimately triumphs.

4. Discussion

The present study shows that ID agents raise the level of cooperation substantially more than II agents, and thereby provide robust support to the importance of neighbor-specific reinforcement learning in solving the prisoner’s dilemma. Similar to [33], our work underlines the contribution of reinforcement learning in achieving higher cooperation.

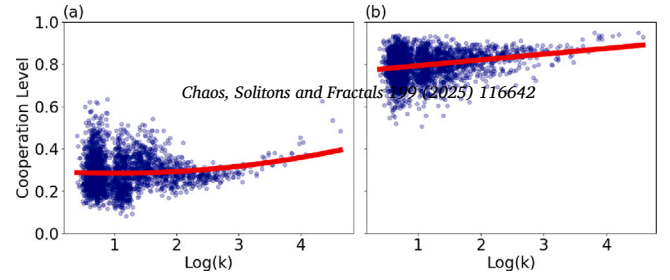


Fig. 11. Relationship between $\log(k)$ and final cooperation level in a scale-free network. To investigate the role of connectivity on cooperation, we examine a reduced scale-free network of 2500 nodes at $D_r = 0.15$. In panel (a), $\rho = 0$ (all II), and we fit a quadratic regression line to capture the non-linear trend between $\log(k)$, the logarithm of node degree, and the resulting final cooperation level. Panel (b) sets $\rho = 1$ (all ID), revealing a strong positive correlation under a linear regression. These outcomes highlight that agents with higher degree tend to cooperate more, although the precise relationship between degree and cooperation depends on whether strategies are aggregated (II) or neighbor-specific (ID).

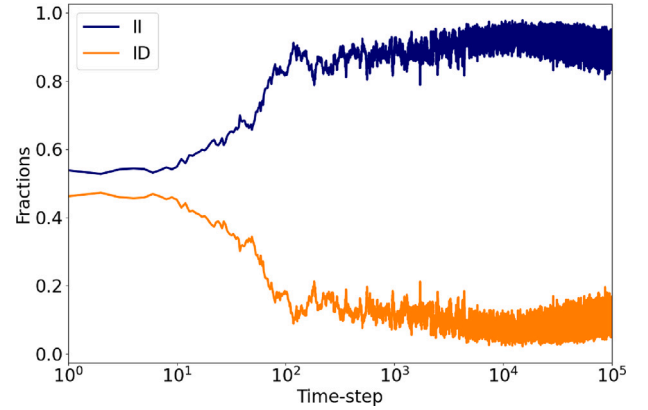


Fig. 12. Temporal evolution of agent proportions in a scale-free network. This figure mirrors the analysis of Fig. 6 but applies it to a scale-free network of 40,000 nodes, focusing on $\rho = 0.5$ and $D_r = 0.15$. Agents initially comprise an equal mix of II and ID types, each updating its strategy by Q-learning and deciding whether to switch learning modes via a Fermi-function comparison of payoffs. Over 10^5 steps, the fraction of II vs. ID agents changes according to the observed rewards, reflecting the capacity of one learning rule to invade the other when advantageous. The plot demonstrates that, in a decentralized scale-free environment, node connectivity and payoff gradients can favor more uniform, stable strategies (II) or foster specialized adaptability (ID), depending on the early distribution of agents and the extent of defection pressure.

However, instead of considering a combination of Q-learning with Fermi updates, we propose to divide agents into two kinds: Interactive Identity and Interactive Diversity. In this regard, ID agents, though keeping different strategies for different neighbors, effectively avoid being exploited while benefiting from mutual interactions. This result confirms the general principle of evolutionary game theory that rich local information, along with adaptive updates of strategies, can stabilize prosocial behaviors even in a harsh or uncertain environment. Indeed, by running several “mini-learners”, ID agents counterbalance the uniform perception that often constrains II agents, especially under intermediate dilemma strengths where the temptation to defect is neither negligible nor overwhelming.

Network structure significantly enhances the benefits of ID learning. Recent work on reinforcement-learning-driven adaptive networks further underscores this point. For example, [43] shows that agents who sever unprofitable links via Q-learning transform an initially regular lattice into a heterogeneous, scale-free graph, thereby amplifying cooperation. Whereas that study lets learning reshape the topology, our work fixes the topology and instead increases the cognitive granularity of Q-learning (II versus ID). Taken together, the two lines of research

indicate that structural plasticity and neighbor-specific learning are complementary mechanisms for sustaining prosocial behavior. Finally, the alternative learning rules surveyed above, including evidential fusion [7], rational conformity [8], weighted-link adaptation [44] and crosstalk-aware memory-one strategies [45], could all be embedded within our II/ID template, opening a rich avenue for unifying disparate strands of human-cooperation research. On grid networks, ID agents tend to completely dominate by forming tight clusters of cooperators which grow over time and outcompete defectors. In scale-free networks, strong heterogeneities are introduced into node degree and influence [46,47]. While the ID agents continue to be generally favorable, high-degree II nodes can anchor the population in partially defective states, thus underlining a competitive tension between ID's granular adaptability and the hub-stabilizing effect. This duality suggests that topological features have to be taken into consideration when systems based on local learning rules are designed or analyzed, including microbial consortia, robotic swarms, and social networks [48–50].

This strength of individual-level adaptation is further emphasized if agents are allowed to evolve their learning type-contingent on payoff-switching between II and ID or vice versa. While ID agents always outcompete II agents on regular lattices by means of homogeneous connectivity that allows them to build local zones of cooperation that eventually spread, clusters of II agents can survive on scale-free networks if they happen to occupy the hub positions initially. Surmounting this structural disadvantage may require ID agents to penetrate the high-degree nodes early, therefore broadcasting cooperative signals to a broad swath of the network. Such patterns reflect real-world observations where different learning or decision-making protocols can coexist and dominant behavioral rules emerge only once local interactions favor an update method [51].

Other important spatiotemporal patterns that arise in the model are related to the problem of cooperation. Namely, ID agents give rise to sharper boundaries between cooperators and defectors, insulating cooperative regions against defecting neighbors, whereas for II agents, the transition is smoother, thus allowing a faster infiltration of defectors, especially if the strength of the dilemma is above a critical value. Indeed, our spatiotemporal pattern analysis shows that ID agents generate well-defined clusters of cooperation in the long run where spatiotemporal snapshots exhibit localized patches of cooperators that prove to be rather impermeable to the encroaching defectors. In contrast, the cooperative regions generated by the II agents are diffuse and quickly get washed away as soon as defection starts to spread. This is because the ID agents can update their Q-values neighbor-by-neighbor, thus responding more precisely to local dangers and reinforcing successful reciprocal relationships at the boundaries of cooperative clusters. For this reason, defection waves spread more slowly through populations of ID agents, explaining why transitions at cluster boundaries are sharper and more stable under the ID condition. These forms of “specialized reciprocity” are in line with the empirical findings in biological and social systems, where often tailored responses to interaction partners solidify mutualistic relations against external disturbances [19,52,53].

Despite these encouraging findings, our model has limitations that open avenues for future improvements. First, we assume that agents incur no additional cognitive cost for processing neighbor-specific strategies, which might be unrealistic for large-scale or resource-constrained systems. Second, the learning rates and discount factors are fixed, whereas real-world agents often adaptively tune these parameters over time. Third, although we explore both grid and scale-free topologies, small-world, random, or dynamically rewiring networks could yield new insights into the interplay between structure and adaptive behavior. Finally, our results remain at the level of computational modeling, so comparing the predictions with empirical or experimental data—such as controlled behavioral studies or multi-robot interactions—would offer valuable validation [54,55].

Looking beyond cooperation per se, there is growing interest in understanding how similar mechanisms drive the emergence of other moral behaviors, including honesty, trust, reciprocity, and punishment. Prior studies have indicated that moral preferences can be closely tied to individual-level decision processes as well as the structural changes in social networks [56,57]. In this vein, future extensions of our reinforcement learning framework could explicitly incorporate moral decision-making, allowing agents to learn not only how to cooperate or defect but also whether to punish, reciprocate, or reward based on evolving personal norms [58–60]. By incorporating adaptive rewiring—where agents can select or sever connections based on moral preferences—one could investigate how moral strategies diffuse, stabilize, or evolve in dynamic network structures. Such expansions would deepen our grasp of the relationship between individual learning, moral norms, and the large-scale propagation of socially beneficial behaviors.

In summary, our findings highlight the transformative impact that granularity in reinforcement learning can have on the emergence and maintenance of cooperation across diverse interaction networks. Whether the focus lies on microbial communities, social collectives, or engineered systems, adopting neighbor-specific adaptation confers a robust defense against exploitation and fosters cohesive prosocial patterns. We anticipate that continuing to refine this approach—through more realistic learning mechanisms, exploration of additional network architectures, and empirical validations—will further bridge the boundaries between evolutionary biology, physics, and computational social science. By broadening the scope to include moral dimensions, we move closer to a comprehensive understanding of how intelligent agents, both natural and artificial, coordinate to achieve collective well-being.

CRedit authorship contribution statement

Hsuan-Wei Lee: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Yi-Ning Weng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no conflicts of interest regarding the content and publication of this manuscript.

Appendix

In Fig. A.1, we offer a diagram for processes based on the definition of the payoff matrix to clarify how the payoffs are computed. The left-hand side of the figure illustrates a focal agent and its four immediate neighbors in a regular lattice structure. For each pairwise interaction between the focal agent and its neighbors, the right-hand side of the figure specifies the corresponding payoffs according to the players' strategies. Specifically, when both agents cooperate (C), each receives a reward $R = 1$. If one cooperates while the other defects (D), the cooperator receives a sucker's payoff $S = -D_c$, while the defector gains a temptation payoff $T = 1 + D_c$. Mutual defection yields the punishment payoff $P = 0$ for both parties. This matrix-based structure governs all agent interactions throughout the simulation.

In Fig. A.2, we present the overall structure of the strategy updating process for both II and ID agents. The process begins with the agent determining its current state. Depending on the exploration–exploitation mechanism, the agent either selects an action randomly (exploration) or consults its Q-table to make a decision (exploitation). In the exploitation path, the agent retrieves the Q-values corresponding to the current state, compares the values for actions C (cooperate) and D (defect), and selects the action with the higher Q-value. If both actions yield identical Q-values, one is selected at random. After

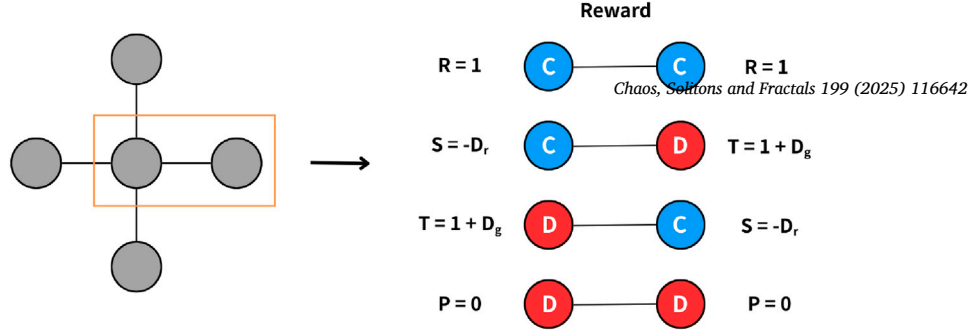


Fig. A.1. Schematic illustration of the reward determination process based on the payoff matrix. The left side shows the focal agent and its local neighborhood in the lattice network. The right side details the corresponding payoff values for each pairwise interaction, determined by the agents' actions (cooperate or defect) and the payoff parameters D_r and D_g .

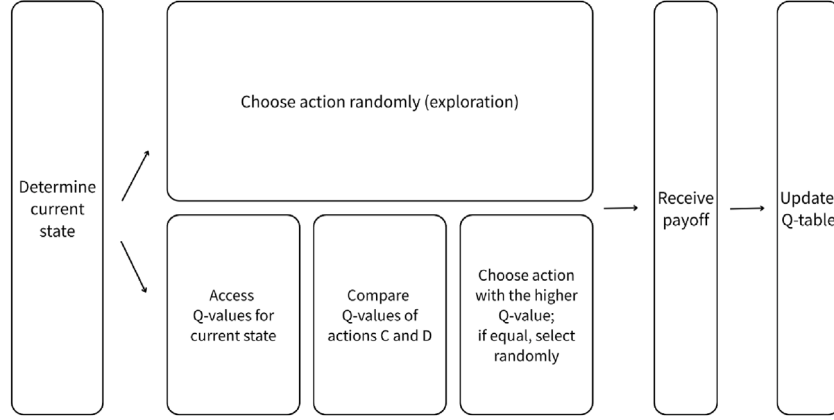


Fig. A.2. Illustration of the overall structure of the strategy updating process for II and ID agents. The flowchart outlines the sequential decision-making steps: determining the current state, choosing an action via exploration or exploitation, receiving the resulting payoff, and updating the Q-table accordingly.

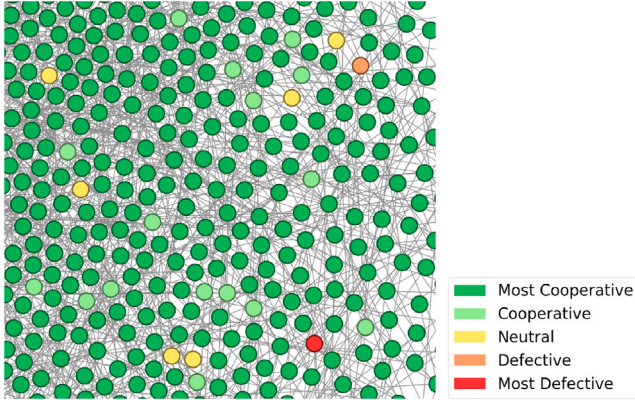


Fig. A.3. Zoomed-in view of the ID agent distribution in the scale-free network shown in Fig. 10(b), with $D_r = 0.15$ and $\rho = 1$. Each node is colored based on the cooperative actions it performs, ranging from “most cooperative” (over 80% of actions are cooperative), “cooperative” (60% to 80% of actions are cooperative), “neutral” (40% to 60% of actions are cooperative), “defective” (20% to 40% of actions are cooperative), or “most defective” (under 20% of actions are cooperative), as determined by neighbor-specific strategies. The image demonstrates the dominance of cooperation among ID agents and underscores the spatial sparsity of defectors.

executing the chosen action and interacting with a neighbor, the agent receives an immediate payoff and updates its Q-table accordingly using the Q-learning update rule.

In Table A.1, we present example Q-tables for two types of agents: Interactive Identity (II) agents and Interactive Diversity (ID) agents. The II agent maintains a larger Q-table of size 2×6 , as its action

Table A.1

Example Q-tables for II (2×6) and ID agents (2×3). When an agent uses its Q-table to select an action, it follows three steps. First, it determines its current state and locates the corresponding column in the Q-table. Second, it compares the Q-values associated with actions C (cooperate) and D (defect). Finally, it chooses the action with the higher Q-value.

(a)						
Action/State	0	1	2	3	4	5
C	0.42	0.15	0.33	0.80	0.27	0.60
D	0.51	0.72	0.44	0.66	0.35	0.20

(a)			
Action/State	0	1	2
C	0.34	0.28	0.55
D	0.62	0.31	0.48

selection depends on six distinct states derived from the combined behaviors of its neighbors. In contrast, the ID agent utilizes a more compact Q-table of size 2×3 , reflecting its design to process each neighbor individually and independently. These Q-tables represent the agents' learned estimations of expected future rewards for choosing either cooperation (C) or defection (D) in a given state. During the decision-making process, each agent first identifies its current state, then retrieves the corresponding column in its Q-table, compares the associated Q-values of C and D, and selects the action with the higher value.

In Fig. A.3, we present a zoomed-in subfigure of Fig. 10(b), which illustrates the distribution of different action states among ID agents on a scale-free network. This visualization offers a closer view of how agents

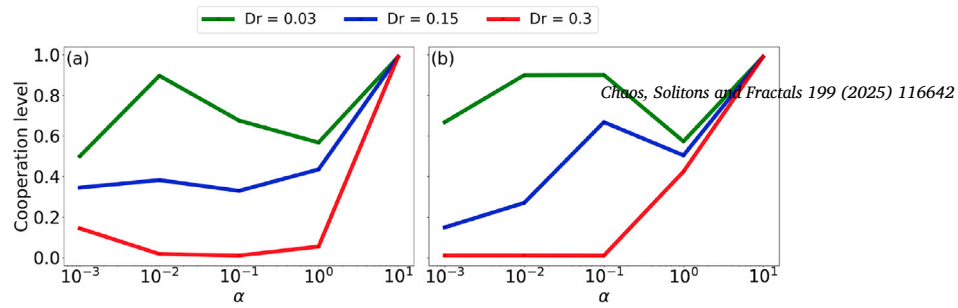


Fig. A.4. Cooperation levels under varying learning rates α , across three dilemma strengths ($D_r = 0.03, 0.15$, and 0.3). Panel (a) shows results for Interactive Identity (II) agents, and panel (b) shows Interactive Diversity (ID) agents. Extreme values of α tend to cause instability or irregular dynamics, while moderate values such as $\alpha = 0.1$ lead to more stable and consistent cooperation. We therefore select $\alpha = 0.1$ as the standard setting throughout our simulations.

adopting neighbor-specific strategies manifest varying levels of cooperativeness. As shown, the population is overwhelmingly composed of cooperative agents, with only sparse and isolated defective nodes. This spatial pattern highlights the ability of ID agents to foster robust cooperation even in highly heterogeneous network environments.

In Fig. A.4, we examine how the learning rate α influences cooperation levels under different dilemma strengths ($D_r = 0.03, 0.15$, and 0.3). Panel (a) displays results for Interactive Identity (II) agents, while panel (b) corresponds to Interactive Diversity (ID) agents. We observe that both excessively small and excessively large values of α tend to induce fluctuations or unstable cooperation outcomes. In contrast, moderate values—particularly $\alpha = 0.1$ —yield relatively smooth and consistent behavior across a wide range of dilemma strengths. For this reason, also following the common practice [33,61], we adopt $\alpha = 0.1$ as the default learning rate in our experiments to ensure both effectiveness and stability.

Data availability

Data will be made available on request.

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